OFFICIAL REFERENCE RESOURCES

Ground Truth Under Attack — Securing the Earth Observation Chain

TSYP 14 Technical Challenge · IEEE GRSS, SPS & CIS Tunisia Section Chapters

Datasets are the official reference data for each track's standard test. Tools are free, verified software that teams can use to work with that data — none are mandatory, but all directly support the deliverables.

Resource	Track	What it is	How we use it in the challenge
SatNOGS Network Observations	Track 1	A crowd-sourced network of ground stations that record and share satellite radio signals.	Analyze a recorded signal and detect digitally added interference or replay.
NASA POWER Daily API	Track 2	A NASA service providing daily weather and solar data: temperature, humidity, wind, solar radiation.	Compare a sensor's readings against a trusted reference to spot inconsistencies.
Copernicus Browser	Track 2 / 3	A platform to browse satellite imagery and data, including the Sentinel missions.	Observe an area and compare how it changes across different dates.
EuroSAT	Track 3	A labeled image dataset built from Sentinel-2, covering forests, crops, residential areas and more.	Use original images, tamper with copies, and test a forgery detector.
gr-satellites	Track 1	An open-source GNU Radio module that decodes telemetry frames from amateur satellites, from a recording or a live SDR.	Decode SatNOGS recordings to recover the original frames before hunting for replayed or altered ones.
Universal Radio Hacker (URH)	Track 1	A complete suite for investigating unknown wireless protocols: demodulation, protocol reverse-engineering, and signal replay.	Inspect a captured signal's protocol structure and simulate a replay attack to test your detector against it.
ruptures	Track 2	A Python library for change-point detection in a signal: sudden bias, gradual drift, regime change.	Automatically detect the step bias and gradual drift injected into the sensor's time series.
PyOD (Python Outlier Detection)	Track 2	A comprehensive Python toolkit bringing together 40+ anomaly-detection algorithms behind one consistent API.	Benchmark several anomaly-detection methods quickly and pick the one that best separates the injected attack from normal readings.
ExifTool	Track 3	The reference tool for reading, writing and inspecting image metadata: EXIF, IPTC, XMP, C2PA.	Check whether an image's provenance metadata was altered or stripped after capture.
C2PA (Content Credentials)	Track 3	An open industry standard (Adobe, Microsoft, BBC, Intel, Arm) for attaching a verifiable history to a piece of media.	Use as the reference model for your provenance chain: what a signed, tamper-evident image record should contain.
Forensically	Track 3	A free browser-based image forensics toolkit: error level analysis, clone detection, metadata extraction.	Get a quick visual read on whether a submitted image shows signs of splicing or resaving before building your own detector.